

TECHNICAL & COMMERCIAL PROPOSAL

Replacement and Refurbishment of Cold Rooms at NGL and Refinery Canteens — Mesaieed

Proposal Ref.	MES/ /2026/CS-170
Date	02 June 2026
To	MANNAI ENERGY
Attention	Mr. Jemina Mani - Junior Engineer-Electrical
End Client	QatarEnergy (NGL & Refinery Canteens, Mesaieed Industrial City)
Subject	Supply, replacement, installation, testing and commissioning of cold-room refrigeration systems — Part 1 (NGL Canteen) and Part 2 (Refinery Canteen).
Validity	15 days from the date of this proposal

1. Introduction

Metri Engineering Services W.L.L. (MES) is pleased to submit this Technical & Commercial Proposal to Engineering Maintenance Company (EMCO) for the replacement and refurbishment of the cold-room refrigeration systems serving the NGL and Refinery Canteens of QatarEnergy at Mesaieed Industrial City.

Our offer has been prepared in full reference to the Scope of Works and Specification (Annexure 1 of Appendix 1) and the associated Schedule of Prices (Annexure 1 of Appendix 2). It covers Part 1 — replacement of the existing walk-in freezer and conversion/replacement of one chiller with a freezer at the NGL Canteen — and Part 2 — turnkey replacement of two central refrigeration rack systems and their rooftop air-cooled condensers at the Refinery Canteen.

MES leverages a proven supply chain for this project: **Bitzer (Germany)** semi-hermetic compressors, factory unit assembly by **Revacco (Turkey)**, and **JMB (Qatar)** 100 mm tongue-and-groove insulated sandwich panels, all selected for reliable continuous operation at the Mesaieed design ambient of up to 52°C.

2. Understanding of Scope

2.1 Part 1 — NGL Canteen

The existing modular walk-in freezer and one modular walk-in fruit chiller — both in service for approximately 20 years — are obsolete and undersized. MES will remove and dispose of both existing units (including refrigeration and electrical systems) and supply, install and commission two (2) new factory-fabricated modular walk-in freezer rooms, each capable of maintaining storage temperatures of -18°C to -22°C and designed for heavy-duty catering use.

2.2 Part 2 — Refinery Canteen

The two existing central refrigeration rack assemblies and their corresponding rooftop air-cooled condensers — also at the end of their service life after approximately 20 years — will be removed and disposed of. MES will supply, install and commission two (2) new factory-assembled refrigeration rack assemblies (one serving the chiller rooms, one serving the freezer rooms) with their associated condensers, controls, oil management, safety devices and interconnecting refrigerant piping, fully integrated with the existing cold rooms and electrical infrastructure.

3. Proposed Technical Solution

3.1 Part 1 — NGL Canteen Walk-In Freezers (x2)

- Two (2) factory-fabricated modular freezer rooms, constructed from JMB (Qatar) prefabricated polyurethane insulated sandwich panels of 100 mm thickness, tongue-and-groove with cam-lock jointing for airtightness; anti-corrosive internal and external finishes; insulated, reinforced floor for heavy trolley and foot traffic. Room dimensions to suit the existing footprints.
- Hinged door with 900 mm clear opening, complete with anti-condensation door/frame heaters, heavy-duty hinges, lock assembly, internal safety release and sealing gaskets.
- Standard accessories: LED lighting, PVC strip curtain, pressure-relief valve and digital temperature display.
- Complete refrigeration system per room: in-room (ceiling-mounted) evaporator with electric defrost and stainless-steel drain pan; external air-cooled condensing unit with Bitzer 4EES-4Y semi-hermetic compressor, integral air-cooled condenser, selected for continuous operation up to 52°C ambient; corresponding evaporator TEA 40 125 8.
- Refrigerant R404A for the freezer systems, with full safety package: digital temperature control, high-/low-pressure protection, oil safety protection and phase-failure relay.

3.2 Part 2 — Refinery Canteen Refrigeration Racks (x2) & Condensers

- **Chiller Rack (Rack 1):** factory-assembled rack with 2 × Bitzer 4EES-4Y semi-hermetic compressors, -10°C Evap / +55°C Condensation duty, remote air-cooled condenser model TUB63 123 2,1, and three (3) corresponding evaporators TEA 40 125 6.
- **Freezer Rack (Rack 2):** factory-assembled rack with 3 × Bitzer 4GE-23Y semi-hermetic compressors, -35°C / +55°C, remote air-cooled condenser model TUB63 233 2_1, and three (3) corresponding evaporators TEA 50 145 8.
- Each rack supplied complete with oil management (oil separator, filters, oil-level regulators and oil-return components), liquid receiver and suction-line accumulator, vibration isolators, refrigerant piping manifolds and ancillary components (filter-driers, moisture indicators, service valves, gauges) as integral factory-assembled items.
- HP/LP pressure switches, transducers, relief valves, gauges and solenoid valves; all calibrated and tested at commissioning.
- Eliwell / AKO digital control panels with safety interlocks
- New factory-assembled rooftop air-cooled condensers rated for sustained operation up to 52°C ambient, with corrosion-resistant frame treatment and coatings suitable for coastal environment, dimensionally matched to the existing rooftop supports (within available footprint).
- Interconnecting refrigerant piping brazed, pressure-tested, evacuated, insulated and recharged; field electrical cabling, terminations and interface to the existing distribution/control panels; full functional testing and commissioning with performance report.

3.3 Equipment Origin

- Compressors — Bitzer, Germany.
- Refrigeration unit / rack assembly — Revacco, Turkey.
- Cold-store sandwich panels — JMB, Qatar (100 mm, tongue-and-groove).
- Controls — Eliwell / AKO digital control panels.

4. Schedule of Prices (SCH-2)

The following lump-sum prices are in Qatari Riyals (QAR) and correspond to Annexure 1 of Appendix 2 — Schedule of Prices.

Ref	Description	Unit	Qty	Unit Price (QAR)	Total Price (QAR)
2.1 Part 1 — Replacement of Existing Walk-In Freezer & Replacement of one Chiller with Freezer (NGL Canteen)					
2.1.1	Removal, dismantling and disposal of existing one (1) Walk-in Chiller, including associated refrigeration and electrical systems.	No	1	10,000	10,000
2.1.2	Removal, dismantling and disposal of existing one (1) Walk-in Freezer, including associated refrigeration and electrical systems.	No	1	10,000	10,000
2.1.3	Supply, delivery and installation of modular Walk-in Freezer Rooms, complete with all accessories, testing, commissioning and handover documentation. Air-Cooled Condensing Unit with: - Compressor Brand: Bitzer / Germany - Compressor Model: 4EES-4Y / Semi-Hermetic - Compressor Quantity: 1 pc per unit - Evaporation / Condensation Temp: -25°C / +55°C - Condenser: Air-cooled (integral / packaged) - Digital Control Panel: Eliwell / AKO - Corresponding Evaporator: TEA 40 125 8	No	2	75,000	150,000
2.2 Part 2 — Replacement of Central Refrigeration Rack Systems & Condensers (Refinery Canteen)					
2.2.1	Removal and disposal of existing Refrigeration Racks and rooftop Air-Cooled Condensers.	LS	2	25,000	50,000
2.2.2	Supply, delivery, installation and commissioning of complete Refrigeration Rack Assembly with integrated remote condenser for the Chiller rooms, complete with all accessories, testing, commissioning and handover documentation. Central Cooling Unit (Compressor Rack) — Chiller Rooms: - Compressor Brand: Bitzer / Germany - Compressor Model: 4EES-4Y / Semi-Hermetic - Compressor Quantity: 2 pcs - Evaporation / Condensation Temp: -10°C / +55°C (Tropical) - Condenser: Remote Air-cooled — Model TUB63 123 2,1 - Digital Control Panel: Eliwell / AKO	LS	1	390,000	390,000

Ref	Description	Unit	Qty	Unit Price (QAR)	Total Price (QAR)
	• Corresponding Evaporator: TEA 40 125 6 (x3 pcs)				
2.2.3	Supply, delivery, installation and commissioning of complete Refrigeration Rack Assembly with integrated remote condenser for the Freezer rooms, complete with all accessories, testing, commissioning and handover documentation. Central Refrigeration Unit — Freezer Rooms: • Compressor Brand: Bitzer / Germany • Compressor Model: 4GE-23Y / Semi-Hermetic • Compressor Quantity: 3 pcs • Evaporation / Condensation Temp: -35°C / +55°C (Tropical) • Condenser: Air-cooled — Model TUB63 233 2_1 • Digital Control Panel: Eliwell / AKO • Corresponding Evaporator: TEA 50 145 8 (x3 pcs)	LS	1	450,000	450,000
TOTAL PRICE OF SCH-2 (QAR)					1,060,000

Total Contract Price: QAR 1,060,000 (Qatari Riyals One Million Sixty Thousand only).

- The price is for the full project; partial award or partial items are not accepted.
- Prices are firm and fixed for the validity period and exclude any taxes or levies that may be applicable.

5. Scope of Inclusions

The lump-sum prices above include the following, for both Part 1 and Part 2:

- Removal, safe refrigerant recovery, dismantling and disposal of the existing chiller, freezer, refrigeration racks and rooftop condensers.
- Supply, delivery, installation, testing and commissioning of all new equipment as described in Section 3 and the Schedule of Prices.
- All refrigerant, lubricating oil, gaskets, sealants, fasteners, insulation and incidental installation/fixing materials directly required to complete the described works.
- Interconnecting refrigerant piping, brazing, pressure/leak testing, evacuation, charge balancing and recharging.
- Field electrical cabling, terminations and interface connections from the existing distribution/control panels to the new equipment (subject to the assumptions in Section 6).
- Minor builder's work directly associated with the equipment removal and installation (equipment pads, supports, sealing of penetrations and making good at affected areas only).

- Method Statements (separate for Part 1 and Part 2), project execution plan, manufacturer datasheets, signed commissioning sheets with warranty certificates, part numbers, list of local suppliers, and final equipment list and system schematic.
- Two (2) year warranty on equipment and workmanship from successful commissioning and handover.

6. Exclusions & Clarifications

Any item, work, material or service not expressly described in this proposal and the priced Schedule of Prices is excluded. Any such item, if subsequently required, will be treated as a variation and priced separately. In particular, the following are excluded:

1. Anything not clearly mentioned in the scope and priced in this proposal.
2. Any safety training costs, PTW permits costs, any training/ any medical tests that has a cost by client.
3. Training of client / end-user personnel and any associated costs.
4. Gate-pass costs and processing, security-pass and access/permit charges for personnel and vehicles.
5. Consultant, specialist and any third-party service costs.
6. Insurance policies of any type (Contractor All-Risk, third-party, workmen's, etc.).
7. Any item carrying a cost impact that is not clearly mentioned in the scope and priced herein.
8. Major civil or structural works, structural strengthening or reinforcement of the building or rooftop supports, and new foundations (beyond equipment housekeeping pads).
9. New electrical infrastructure — new MDBs/SMDBs, transformers, upgrade of the incoming supply, or new cable containment beyond connection to the existing panels; adequate spare capacity at the existing boards is assumed.
10. BMS hardware, software, licences and head-end integration beyond the standard volt-free signal interface of the supplied controllers.
11. Provision of standby or temporary refrigeration during the shutdown period, unless separately instructed.
12. Handling or disposal of hazardous materials (e.g. asbestos), if encountered.
13. Statutory authority approvals, third-party inspection/certification and associated fees.
14. Special scaffolding or craneage beyond standard lifting equipment, and any out-of-hours / night-work premiums, unless separately instructed.
15. Price escalation arising from currency fluctuation, customs/duty changes, or programme extension beyond the validity period.
16. Works at any location other than the NGL and Refinery Canteens, Mesaieed.

Technical clarifications:

- New freezer-room dimensions for Part 1 are based on the existing footprints and will be confirmed at site survey.
- Part 1 condensing units are offered as external air-cooled packaged units (compressor with integral condenser) located outside the cold rooms; final location to be agreed at site.

- Equipment capacities are based on the supplier's selection for 52°C ambient operation and are subject to confirmation against the detailed cold-room heat loads at design stage.
- Refrigerant R404A is offered for the freezer systems in line with the specification; a lower-GWP alternative can be offered subject to QatarEnergy approval.
- Control is provided via Eliwell / AKO microprocessor panels with safety interlocks and alarms; full PLC / BMS integration is excluded unless separately specified.

7. Commercial Terms

- **Currency:** Qatari Riyals (QAR), exclusive of any applicable taxes or levies.
- **Contract price:** QAR 1,060,000 lump sum for the full project (no partial items).
- **Price validity:** 15 days from the date of this proposal.
- **Payment terms (to be agreed):** 50% advance against Purchase Order; 35% on delivery of equipment to site; 10% on completion of installation and commissioning; 5% on final handover.
- **Completion period:** within twelve (12) calendar months from the effective date, inclusive of procurement, installation, testing, commissioning and handover; detailed project execution plan within fourteen (14) days of award.
- **Warranty:** two (2) years on equipment and workmanship from successful commissioning and handover.

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